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A tree detection method based on trunk point cloud section in dense plantation forest using drone LiDAR data

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ABSTRACT

Single-tree detection is one of the main research topics in quantifying the structural properties of forests. Drone LiDAR systems and terrestrial laser scanning systems produce high-density point clouds that offer great promise for forest inventories in limited areas. However, most studies have focused on the upper canopy layer and neglected the lower forest structure. This paper describes an innovative tree detection method using drone LiDAR data from a new perspective of the under-canopy structure. This method relies on trunk point clouds, with under-canopy sections split into heights ranging from 1 to 7 m, which were processed and compared, to determine a suitable height threshold to detect trees. The method was tested in a dense cedar plantation forest in the Aichi Prefecture, Japan, which has a stem density of 1140 stems·ha^{-1} and an average tree age of 42 years. Dense point cloud data were generated from the drone LiDAR system and terrestrial laser scanning with an average point density of 5000 and 6500 points·m^{-2}, respectively. Tree detection was achieved by drawing point-cloud section projections of tree trunks at different heights and calculating the center coordinates. The results show that this trunk-section-based method significantly reduces the difficulty of tree detection in dense plantation forests with high accuracy ($F1 - \mathrm{Score} = 0.9395$). This method can be extended to different forest scenarios or conditions by changing section parameters.

1. Introduction

Drone remote sensing is an important part of modern forest surveys. The accelerated development of drones and remote sensing has provided new options for forest surveys, including increased flight time and ability to carry relatively heavy payloads. Miniaturization of Inertial Measurement Units (IMUs), global navigation satellite system (GNSS) receivers, Light Detection and Ranging (LiDAR) and advancement of drone systems have contributed to the construction and maturation of drone LiDAR systems (DLS) that are capable of autonomously collecting high point densities (Kaartinen et al., 2012; Wang et al., 2016; Kukkonen et al., 2021). Another common method for forest inventory is terrestrial laser scanning (TLS) (Liang et al., 2016, 2018; Holmgren et al., 2019; de Paula Pires et al., 2022). Compared with DLS, which focuses on the canopy by detecting from the above layer, TLS sets up scanning stations on the ground and features detailed mapping of the trunk and under-canopy information. Therefore, TLS can provide a much more valuable under-canopy point cloud dataset for studies of forests that require information about the under-canopy layers (Polewski et al., 2016; Dai et al., 2019).

Sustainable forest management and forestry policy development are predicated on understanding the current state of the forests. The forestry sector increasingly demands accurate three-dimensional (3D) information on forest structure (Chang et al., 2013; Dersch et al., 2021). Forest structure analysis requires information on tree attributes, such as species, tree height, diameter at breast height (DBH), canopy openness, tree density, and spatial distribution. Specifically, tree density and height are used to assess the forest structure and crowding of standing trees by calculating the relative yield and estimating the evapotranspiration of the entire forest system (Komatsu and Kume, 2020). Traditional field surveys allow tree-to-tree measurements of small plots to obtain detailed and accurate forest information; however, large-scale manual surveys are time-consuming and laborious (Hyyppä et al., 2020). LiDAR-based remote sensing methods produce continuous dense point clouds to characterize tree structures in a 3D space (Fernández-Guisuraga et al.,

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Abbreviations

IMUs	Inertial Measurement Units
GNSS	Global navigation satellite system
DLS	Drone lidar systems
TLS	Terrestrial laser scanning
CHM	Canopy-based height models
DBSCAN	Density-based spatial clustering of applications with noise
OWL	Optical wood ledger

2022). Further analytical processing of forest point clouds provides new ideas for obtaining the required tree attributes at a reasonable cost over a wide area.

Single tree detection methods can be roughly divided into two main categories: canopy-based height models (CHM) using canopy raster as the basic information and point cloud-based direct manipulation methods (Hyyppä et al., 2001; Zhen et al., 2016; Hao et al., 2021). The canopy-based approach inverts the canopy height model and simulates the filling process using a sufficient number of pixels. As water fills the basins and joins water from adjacent basins, watershed edges are established (Tang et al., 2007; Ene et al., 2012). This method is sensitive to the resolution and amount of smoothing used to produce the input canopy height (McGaughey, 2020). The point cloud-based approach offers alternative ideas, significant local maxima within the threshold window are selected as seed points. The *k*-means clustering method is an iterative partitioning method that uses distance criteria for seed points to group LiDAR points into a set of clusters (Gupta et al., 2010). Density-based spatial clustering of applications with noise (DBSCAN) method allows to divide regions with sufficient density into clusters and efficiently filter out regions with low density points, eventually finding the shape of the trunk (Wang et al., 2019; Husain and Chandra Vaishya, 2020; Zhou et al., 2022). Wang et al. (2008) created a distribution map of canopy reflections to infer the location of the canopy by scanning the tree voxels from top layer to ground. Also a few studies have also extracted complex surface feature units, such as stem curves, as the basis for tree recognition (Jaskierniak et al., 2021). The dense point clouds acquired by DLS and TLS provide a more detailed reconstruction of tree structures, including the crown and trunk (Dersch et al., 2021). High-density trunk points make it possible to detect tree positions based on the trunk center coordinates. The bottom-up method does not depend on canopy detection and performs well in detecting scenarios with unclear canopy boundaries in dense forests.

Abandoned plantations were characterized by high stem density and trees under the same growth conditions. Owing to the lack of management and thinning, the dense canopy overlaps and renders boundary identification difficult, which prevents understory vegetation from obtaining sunlight. The sparse understory vegetation, bare forest floor, and lack of multi-stemmed trees create a relatively simple under-canopy structure. These characteristics enable trunk analysis based on the under-canopy trunk information.

This paper describes an innovative tree detection method based on the high-density trunk point cloud under a canopy using drone LiDAR data. This method relies on trunk point clouds, and after sectioning and projecting the under-canopy point cloud, the center coordinates of the trunk polygon are used as the tree position. This is then compared with the TLS dataset to determine an appropriate trunk processing parameter that is then used for tree detection.

The objectives of this study were to: (1) reconstruct point cloud models with multi-layered forest structure (upper canopy and understory) using TLS and DLS, (2) create under-canopy section datasets from DLS with different height thresholds and projection parameters and compare tree detection results, (3) determine the appropriate section parameters for DLS datasets and provide a tree detection method applicable to dense plantation forests. Our next goal is to perform this tree detection method over a large area using DLS data.

2. Materials and methods

2.1. Study area

This study was conducted in a plantation forest with an area of 1.1 ha, located in Wagocho, Aichi Prefecture, Japan (35°01′41″ N, 137°22′24″ E) (Fig. 1). It consists of Japanese cedar (*Cryptomeria japonica* D. Don), planted in 1980, and broadleaved tree species (*Quercus acutissima* Carruth.). The mean DBH, height, and density of the stand was 31.3 cm, 17.3 m, and 1140 stems·ha^{-1}, respectively.

2.2. Terrestrial laser scanning data

The TLS data were captured using an optical wood ledger (OWL) laser scanner (Adin Inc., Tokyo, Japan) set to capture Level 2 quality scans (11.3 mm point spacing at 30 m). It captured 51 scanning stations, ~10 m apart, as well as the surrounding trees. The point density of the scanner was ~6500 points·m^{-2}. The scan results were co-registered in OWL ManagerViewer 1.2 (https://www.owl-sys.com/) using scan-to-scan matching. The quality of the matches was assessed using manually adjusted scans with common features found within the point cloud. To geo-register the TLS data, the approach discussed in Hillman et al. (2021) was implemented. Using a minimum of six common features identified in the geo-rectified drone LiDAR point clouds (such as tree stems and stumps), the relevant translation and rotation were applied to the TLS point cloud. Visual inspection of the point clouds was conducted to ensure that the registration results were aligned with the DLS.

2.3. Drone LiDAR data

Drone LiDAR data were collected on November 23, 2021, using a Zenmuse L1 laser scanning system installed in an enterprise drone Matrice 300 RTK quadcopter (DJI Inc., Shenzhen, China). Zenmuse L1 integrates a Livox LiDAR module, high-accuracy IMU, and camera with a 1-inch CMOS on a 3-axis stabilized gimbal. The flight altitude was set to 50 m relative to the ground using the terrain function. The payload settings were triple echo mode, 160 kHz LiDAR sample rate, and non-repeat scan mode. The scanner was set to face the ground at 90°, and the laser side overlap was set to 90%. This setup maximized the penetration of laser pulses through the upper canopy to the ground, generating high-density point clouds under the canopy.

2.4. Data processing

2.4.1. Point cloud generation and normalization

The post-processing of point cloud data captured by Zenmuse L1 was conducted in DJI Terra 3.3 (https://www.dji.com/). LiDAR point cloud processing includes calculating POS data, fusing point clouds and visible light data, and exporting point clouds as .LAS standard format. The base station data corresponding to the point cloud data from the D-RTK 2 mobile station were used to geo-reference the point coordinates. Point cloud density was set as high density (referring to the original sampling rate), which uses 100% of the data for processing.

Point cloud merging of multiple scanning stations from OWL was performed in OWL ManagerViewer 1.2 (https://www.owl-sys.com/). The registered data were subjected to a closed-source trunk curve detection algorithm (commercial software that accompanied the laser scanner) to detect trees and calculate the DBH. The identified trees were mapped as circles with the DBH size as a reference on a two-dimensional (2D) map. Tree Walkthrough software was used to view the 3D point cloud data. Detection results were marked as white ring bands on the trunk point cloud, with notes on tree ID, DBH, and tree height.

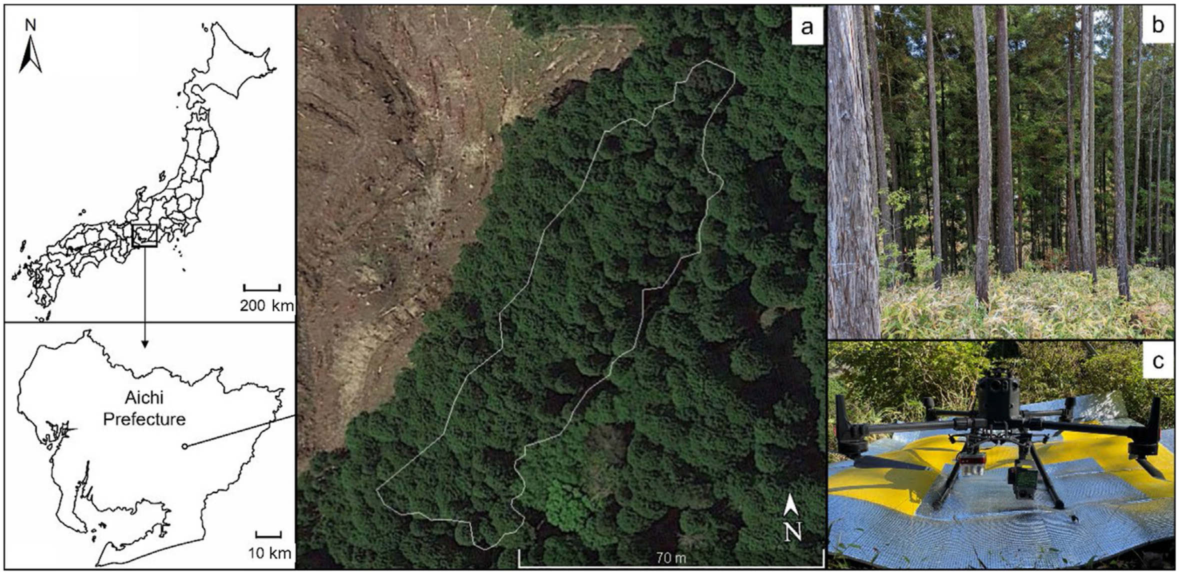

Fig. 1. The study area in Aichi Prefecture, Japan. (**a**) Location of the plantation forest and field plot. (**b**) Forest perimeter and interior views. (**c**) Dji Matrice 300 with Zenmuse L1 laser scanning system.

Ground point cloud classification and extraction are required to generate a continuous ground surface for digital elevation modeling (DEM). Both DLS and TLS point clouds were categorized as ground and off-ground points using the cloth simulation filtering (CSF) method, similar to that of Zhang et al. (2016). The original point cloud was inverted and clothed. Evaluating the interactions between the cloth nodes and corresponding LiDAR data can determine the final cloth shape. Ground points were then extracted from the point clouds by comparing the original points with the generated surface (Zhang et al., 2016). The CSF algorithm was run in CloudCompare v.2.11.3 (http://www.danielgm.net/cc/) on a high-performance computer with a 12-core processor. The main advanced classification parameters for the plantation sites were: cloth resolution = 0.1 m, maximum iterations = 500, and classification threshold = 0.1 m (Zhang et al., 2022). Each off-ground point

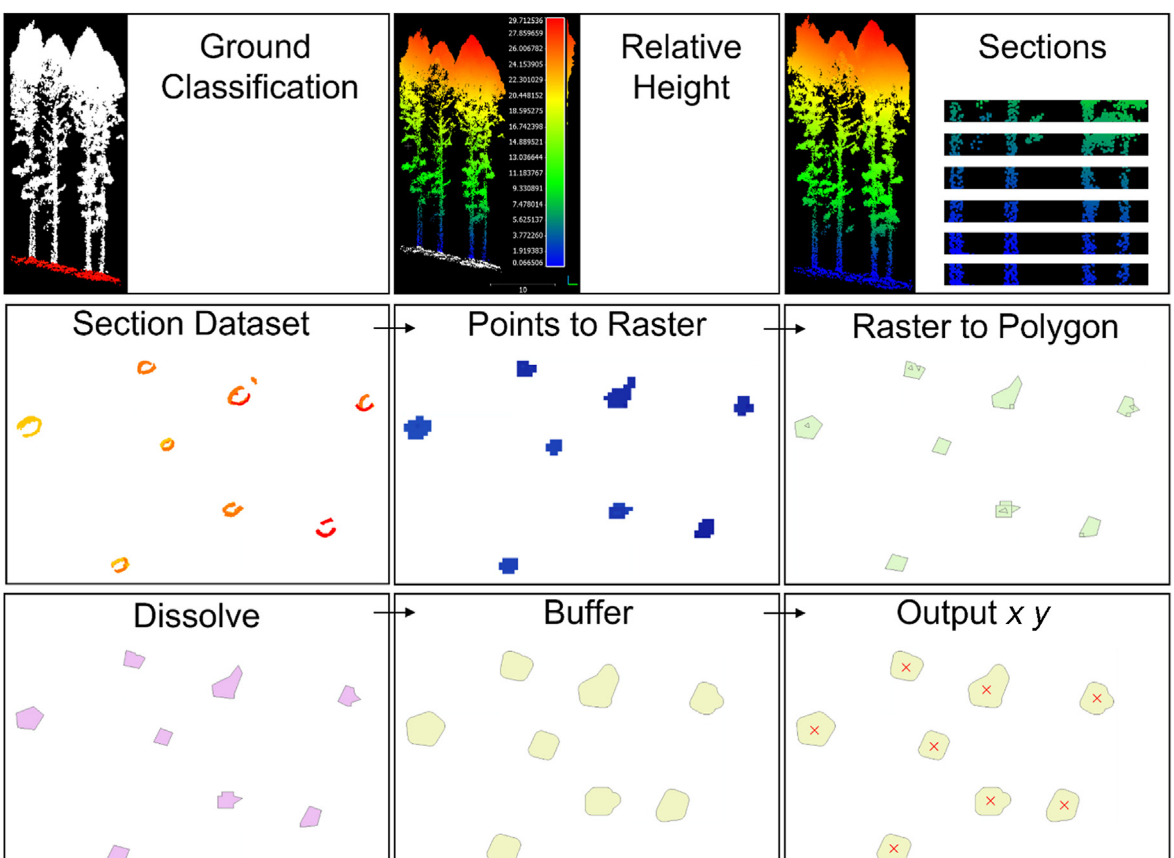

Fig. 2. Generalized flowchart for this study. The three operations of the first line were completed in CloudComapre, and the rest were performed in ArcGIS.

was normalized to the ground by computing the distance between the off-ground and ground clouds.

2.4.2. Under-canopy section processing

The normalized point cloud set was first used to extract the under-canopy point cloud containing the tree trunk, based on the height threshold. Growth conditions, canopy base height, and structural characteristics of Japanese cedar were also considered. To obtain a higher proportion of trunk parts, the high canopy and low understory vegetation points were removed. Point clouds of the under-canopy between 1 and 7 m were extracted at 1-m intervals (relative to ground level). This produced six sets of point cloud sections with different heights, using a filter by value function for DLS point cloud data. These point cloud section datasets were discrete and irregular under-canopy point clouds. They were used as input files for further processing to obtain projection areas that were independent of each other externally, but continuous and complete internally, and could be used to represent the tree trunk position.

2.4.3. Trunk detection

To obtain the projection area of the tree trunk, the section dataset was further processed in ArcMap 10.8.1 (https://desktop.arcgis.com/en/arcmap/) (Fig. 2). Firstly, the *LAS dataset to raster* tool was used to create a raster using elevation values stored in the LiDAR points, referenced by the LAS dataset. This dataset uses binning and selects an average cell assignment method to determine each output cell by using the points that fall within its extent. Additionally, it uses the simple void fill method to average the values from data cells to eliminate small gaps where there is no data. Secondly, the *Raster to Polygon* tool was used to convert the raster dataset to simplified polygons. This was done by setting the polygon's line segments to a minimum, while smoothing and keeping the data as close as possible to the original raster edges. Thirdly, the *Dissolve* tool was used to create and aggregate continuous polygon regions by eliminating the boundaries between elements with the same properties. These operations were performed to ensure smooth outside boundaries and remove the inside boundaries of the trunk polygon. The above-canopy drone LiDAR measurement method requires laser pulses to penetrate the canopy twice (transmission and reception) to ensure that an approximately circular trunk projection area can be generated, even if the trunk point cloud is incomplete, for example, if the point cloud was missing or only partially reconstructed. A *buffer fill* tool was then used to create buffer polygons around the original trunk polygons to enable filling of the interior of the hollow trunk polygon. The excess buffer area will incorrectly connect areas with different polygon properties (trunk adhesion). Conversely, an insufficient distance will not provide the filling effect. Considering DBH, a linear distance of 20 cm was set. Fourthly, a function for projection area calculation was added to calculate each individual projection area and compare it with the set threshold. This removes the problematic areas with the expanded projection area or the under-canopy vegetation with a discrete projection area that is too small. Finally, a dataset consisting of the projection area of each tree trunk was obtained, as independent polygon features. The position of the tree was represented by calculating the *x*- and *y*-coordinates of the centroid. This operation was implemented in the attribute table of the feature layer using the *Calculate Geometry tool*.

The reference detected result was established from the TLS point cloud dataset with commercial software that accompanied the laser scanner. It was then manually corrected ensuring the accuracy of the detected number and tree positions (Fig. 3). The trunk detection results from the 7-section dataset of DLS were compared with the standard results, based on the detected number, and the percentage error and $F1-\text{score}$ were used to evaluate the accuracy, as shown below in Equations (1)–(4).

$$\delta = \left| \frac{v_A - v_E}{v_E} \right| \times 100\% \tag{1}$$

where δ is percent error, v_A is actual value observed, v_E is expected value.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$F1 - \text{Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

where TP is True Positive (correctly detected), FP is False Positive (falsely detected trees), FN is False Negative (trees not detected).

3. Results

3.1. Under-canopy reconstruction

In this study, drone LiDAR and terrestrial LiDAR aided in a successful reconstruction of multi-layered forest structure models by setting reasonable flight parameters and a high density of scanning stations. Both datasets had high-precision geographic coordinates and aligned well. Fig. 4a shows the both TLS and DLS registration results at a study-area level. Fig. 4b shows the registration details of the tree stems. Fig. 4c shows details of the tree registration results from the southern side of the study area. The experimental results showed that both datasets reconstructed the forest structure, including lower vegetation, with high spatial accuracy. The accuracy of the reconstructed under-canopy structure was assessed, as it greatly affects the intactness of the trunk portion. The point cloud density of the under-canopy structure was calculated,

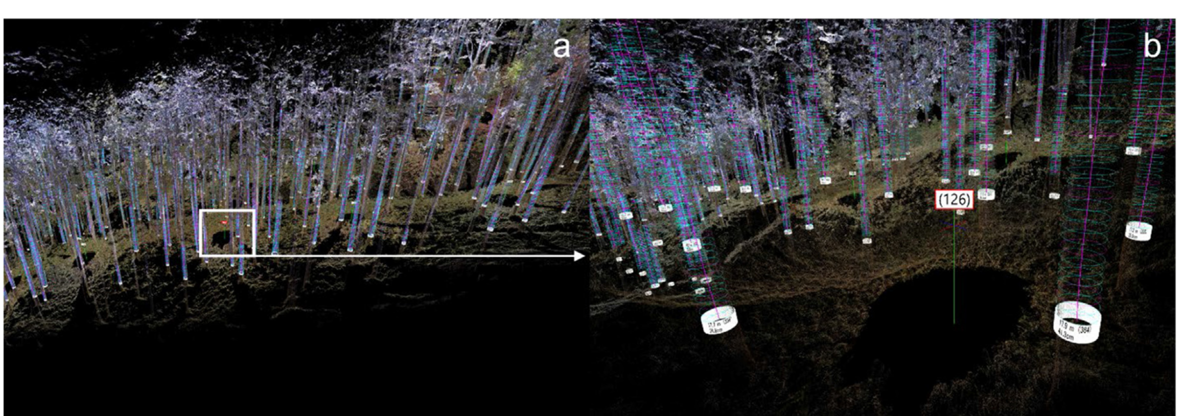

Fig. 3. Tree detected result for OWL (TLS) data. (**a**) 3D point cloud data view in *Tree Walkthrough*. (**b**) Tree information marks, include tree ID, DBH and tree height.

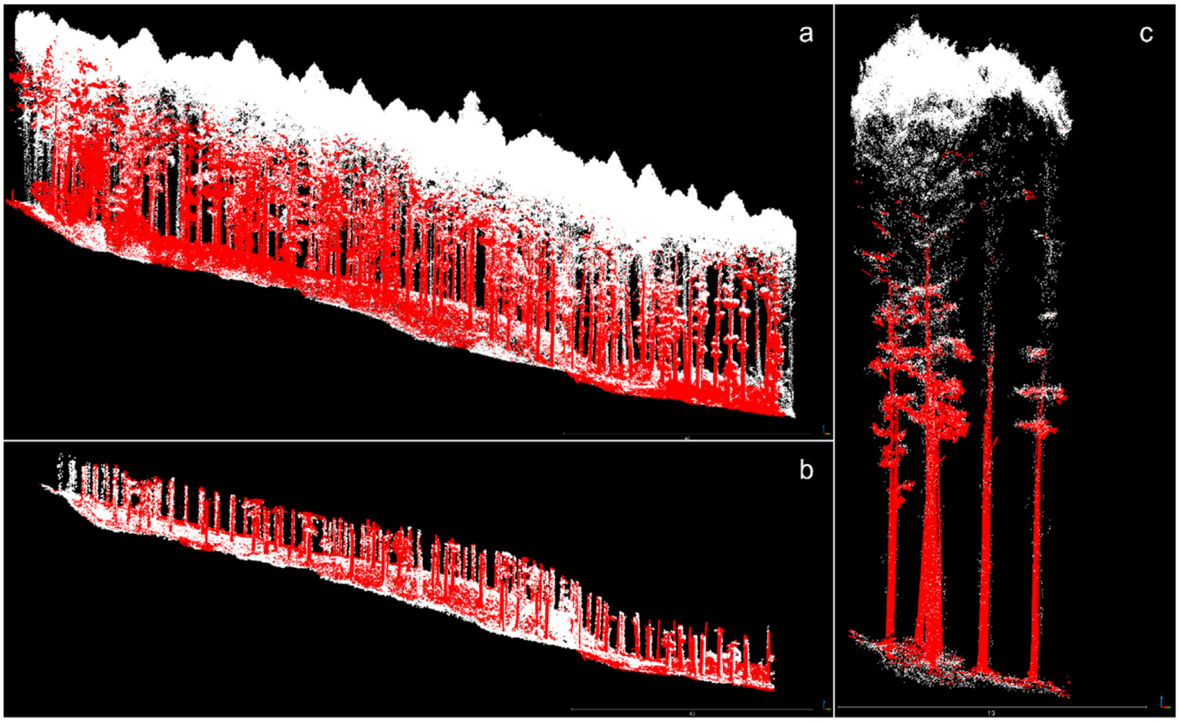

Fig. 4. Registration results for TLS and DLS datasets. The red color represented the TLS point clouds and the white color represented DLS point clouds. (**a**) Overall view registration result. (**b**) Tree trunks registration details. (**c**) Concentrated and zoomed-in registration result from trees in the southern area of the study area. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

and the ratio of under-canopy point cloud to total point cloud was used as an evaluation index. Below 7 m was classified as the under-canopy forest point cloud, which is the range for creating the section dataset.

First, both DLS and TLS reconstructed high-density multi-layered forest structures with an average point cloud density of 5032.63 and 6593.40 points·m^{-2}, respectively, as shown in Table 1. The proportion of under-canopy point cloud to all point clouds were 13.94% and 76.33%, respectively. The terrestrial LiDAR scanning system demonstrated significant advantages for reconstructing under-canopy structures. Starting at a height of 1 m, analysis of the DLS section data indicated that as the section height gradually grew (by 1-m intervals), the proportion of each height level first decreased and then increased, reaching a minimum at the 2–3 m section layer. The point cloud density was only 13.42 points·m^{-2}, accounting for 0.36% of the total point cloud.

3.2. Trunk sections comparison

In the entire study area, the reference dataset was obtained using

Table 1
Summary statistics of point density and proportion of understory point cloud to all point cloud for each section height, point cloud derived from TLS and DLS measurements.

Measurement	Trunk section height (m)	Points	Density (m^2)	Proportion (%)
TLS	All	61202552	6593.40	
	<7 m	46714830	5032.63	76.33
DLS	All	34760222	3744.75	
	<7 m	4844287	521.88	13.94
	6–7 m	141773	15.27	0.41
	5–6 m	124533	13.42	0.36
	4–5 m	142732	15.38	0.41
	3–4 m	151274	16.30	0.44
	2–3 m	154252	16.62	0.44
	1–2 m	168304	18.13	0.48

commercial software that accompanied the laser scanner (Optical Wood Ledger, Adin Inc., Tokyo, Japan) and manually corrected; a total of 761 trees were detected of whole survey site. The cedar planted area of 2905.69 m^2 was in the center of the study area near the west. The TLS initial results showed that 181 trees were identified, and after manual removal of incorrect results, a total of 166 trees with a density of 645.5 stems·ha^{-1} were correctly detected in the cedar plantation area (Fig. 7). This reference result was compared to the DLS section detection results by performed overlay operations.

When using only DLS point cloud datasets of the six different layers (from 1 to 7 m) as the input, the detection results showed a decrease and then an increase with increasing layer height. This is similar to the previous trend of the proportion of section point clouds to the total point cloud for each layer. The minimum detected number was found in the 4–5 m height section dataset with 181 trees, which was slightly higher than the actual value, and a percent error of 9.03% with $F1-\text{Score}$ at 0.9395. The dataset with the most significant percent error occurred in the lowest height section of 1–2 m, with 200 trees detected, and a percent error of 20.48%, while the remaining four sets all had detection errors below 18.67% (Fig. 5). The high $F1-\text{Score}$ appeared in the data with a height of 3–4 and 4–5 m at 0.9188 and 0.9395, the lowest with 6–7 m at 0.8958 (Table 2).

Furthermore, all DLS tree detected results were visualized by section height and overlaid on the original under-canopy point cloud to analyze reasons for the errors. If the detection results match the actual trees, the outcome points will overlap at the section height of the trunk point cloud. First, all other datasets overestimated the results, with the highest overestimation in the 1–2 m dataset, which had some understory vegetation point clouds. Despite designing the projection area threshold to filter out vegetation with a small projection area, there were points where the projection areas were still larger than the minimum threshold. Therefore, there was still a small amount of understory vegetation that was not removed (Fig. 6a) and caused a percentage error of 20.48%, include 0 omission error and 34 commission errors, $F1-\text{Score}$ at 0.9071. In the 2–3-m range dataset, the result was still slightly overestimated,

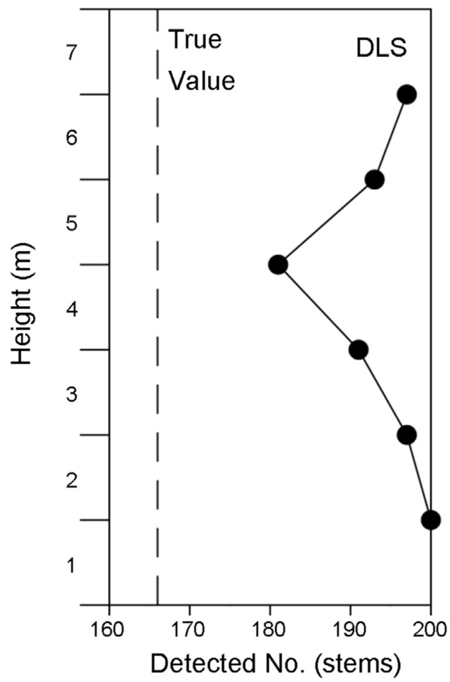

Fig. 5. Relationship between section height and detected number.

Table 2
Detection results, detailed error composition and $F1-\text{Score}$ for 6 height sections (FN: False Negative, trees not detected; FP: False Positive, falsely detected trees; TP: True Positive, correctly detected).

Trunk section height (m)	Detected No.(stems)	Omission errors (FN)	Commission errors (FP)	Correctly detected (TP)	$F1-$ Score
6–7	189	7	30	159	0.8958
5–6	193	4	31	162	0.9025
4–5	181	3	18	163	0.9395
3–4	191	2	27	164	0.9188
2–3	197	2	33	164	0.9036
1–2	200	0	34	166	0.9071

with a percentage error of 18.67%. As the understory vegetation height decreased, the projected area could not be detected, as tree trunks also decreased. With further increases in section layer height, the percent errors for 3–4 and 5–6 m were 15.06% and 16.26%, respectively, providing acceptable accuracy with $F1-\text{Score}$ at 0.9188 and 0.9025 (Fig. 6a and b). Almost all detected points matched the trunk point cloud. Above 5 m, the detected results showed larger commission errors with increasing height. The overlay data of the original under-canopy point cloud showed some overestimated trunk points attached to branches. This suggests that as height increased, branches growing at lower heights were included in the section dataset caused increase in omission errors. Additionally, the distance parameters used by the *buffer fill* and *dissolve* tools were insufficient to fill and merge the trunk and branch projections, which were eventually incorrectly detected as trunks with separate projection areas and also caused the higher commission errors (Fig. 6c).

3.3. Trunk spatial distribution map

The spatial distribution of the seven datasets were plotted, including the standard results, according to the x and y coordinates from spatial distribution analysis (Fig. 7). The dataset with the highest $F1-\text{Score}$ (3–4 and 4–5 m) did not precisely match the standard results, and there were 27 and 18 false detections in the study area. For the dataset with low $F1-\text{Score}$ (1–2 m), many overestimations occurred, and commission errors were not overlapped with other datasets. This indicated that there was a large amount of understory vegetation in this height interval, and that the amount of understory vegetation above this interval was significantly reduced. Most of the detection results for the datasets of 5–6 and 6–7 m were similar to the reference data, with only a small range of deviations in position. This was the effect of the trunk and branches being filled and dissolved into the entire project area, which then shifted the center coordinates. Moreover, the two sets of detection results generally overlap.

4. Discussion

4.1. Results comparison

The aim of this study was to develop a tree detection method, based on high-density trunk point clouds under the canopy only from DLS data. This study indicates that trunk sections with sufficiently dense point clouds (1 m in thickness, point cloud density $\geq$15 points$\cdot$m^{-2}) support the generation of regular trunk projection polygons using operations such as projection and filling. Terrestrial laser scanning is widely used to scan tree structures and generate high-density point clouds, which have more than 5000 points$\cdot$m^{-2} (Jacobs et al., 2021). However, drone with multi-echo LiDAR scanner and flown at low altitudes of 50 m were sufficient to meet the requirements of trunk point cloud density of the method adopted in the present study. An unexpected finding was that the section data for each height overestimated the number of trees to various

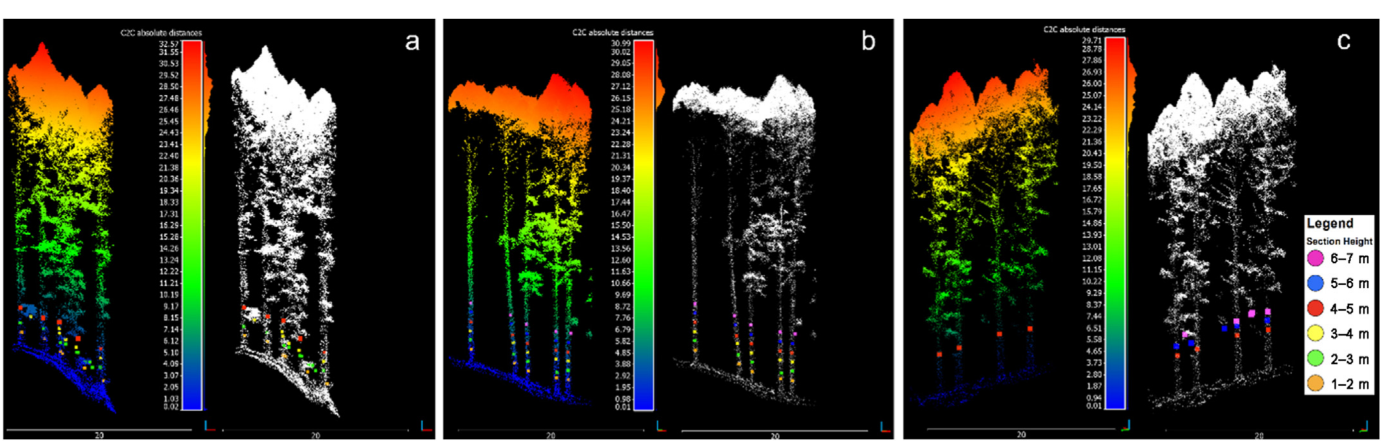

Fig. 6. Visualization of 3D point cloud and all DLS tree detected results by section height. (**a**) Section heights below 5 m and affected by understory vegetation. (**b**) Ideal trunk detection scenario without understory and branch interference. (**c**) Section heights above 5 m and affected by branches.

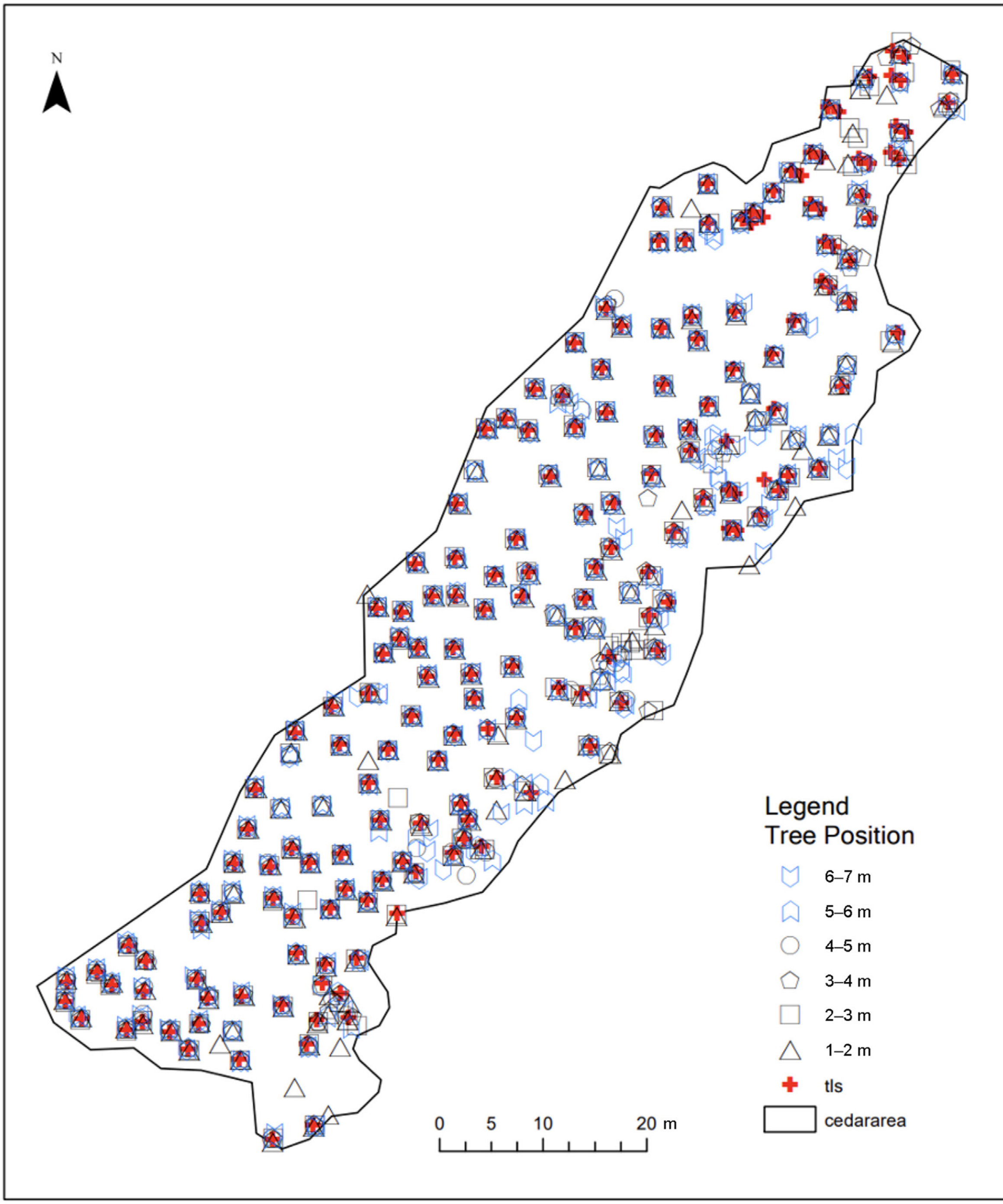

Fig. 7. Spatial distribution of detection results for different section heights, including reference data manually corrected from OWL (TLS).

levels, with the percentage error ranging between 9.03% and 20.48% and $F1 - \mathrm{Score}$ from 0.9189 to 0.9482. Both excessive high and low section heights lead to the increase in commission errors (30 errors at 1–2 m section and 20 at 6–7 m section). Omission error was increasing with height, 0 error at 1–2 m section and 4 errors at 6–7 m section. These results still have acceptable accuracy compared with the previous results of studies by Ferreira et al. (2020) and Forzieri et al. (2009) who obtained higher accuracy using deep learning and morphological operations.

4.2. Comparison with DBSCAN algorithm

Fu et al. (2022) achieved high accuracy in the tree segmentation of plantation and mixed forests with the density-based spatial clustering of applications with noise (DBSCAN) algorithm and completely using the features of the natural separation between tree trunks. We also run the DBSCAN algorithm using the DLS dataset in this paper, in order to obtain comparable results for comparison with the proposed method (Fig. S1). Three different sets of forest structures from the DLS were used as input data: overall point clouds containing the canopy, point clouds of the complete trunk from 1 to 7 m, and point clouds of 4–5 m trunk section. The overall result showed that all canopy parts were classified into the same cluster and the trunk parts are not correctly identified, which may be caused by the high-density stand conditions where there were no specific divisions and gaps between canopies and canopy point clouds were continuous together. The difference in point cloud density of trunk between DLS and TLS is also difficult to ignore. In contrast to the previous study where TLS data was used as input and the scanner scanned the tree trunk directly from the ground, the different canopy thickness during DLS measurement caused the change in penetration of laser pulses and finally led to the high discrepancy of the point cloud of the trunk parts. The trunk dataset after removal of the canopy, both as a whole and in section, showed acceptable accuracy in the results with $F1 - \mathrm{Score}$ at 0.9432 and 0.9538, respectively. A divisional automatic DBSCAN algorithm was utilized for selecting multiple Eps in terms of different objects, understory and trunk (Gaonkar and Sawant, 2013; Fu et al., 2022). The highest Eps level ($\mathrm{Eps} = 2.15$, $\mathrm{MinPts} = 480$) clustered understory vegetation and low branch foliage well when the complete trunk was used as data input. Although the point clouds distribution of tree trunks has continuity in the z-axis, the irregular variation of density led to complicated parameter settings, and the same trunk could be clustered into multiple parts, eventually leading to overestimation of results (FP = 20). In particular, for section data with the following parameters applied, $\mathrm{Eps} = 0.6$, $\mathrm{MinPts} = 3$, the point clouds have 181 clusters for trunk section, understory vegetation. For the trunk clusters, a total of 165 trees were correctly detected in total 180 detected number, 15 commission errors and 1 omission error. In general, DBSCAN achieved higher accuracy results ($0.9395 < 0.9432$, 0.9538) using proven algorithms and appropriate parameters, but the process of parameter determination usually took considerable time. The proposed method required the *buffer fill* tool parameter based on the DBH measured in the field and provided similar accuracy.

4.3. Tree height analysis

The trunk-based detection method can also be used to calculate tree height by extracting CHM values from the tree position points as the detected trunk centroid usually corresponds to the center point of the tree crown. In the dataset of 4–5 m section, the tree height was extracted using the CHM at the location of the trunk centroid. The tree heights obtained using TLS were generally lower than those obtained using DLS. This difference can be attributed to the scanning method of TLS. This method had difficulty reconstructing the point cloud of the upper canopy layer by scanning from the ground, and the single radar echoes were difficult to penetrate and reflect. The lack of an upper canopy structure led to a lower tree height than that of the drone LiDAR. The tree height scatter plot shows that for the number of trees above 22 m, TLS was significantly less than that of drone LiDAR (Fig. S2). Generally, the average tree height of DLS was 32.4% higher than that of TLS (26.54 and 20.04 m, respectively). This may be attributed to the dense leaf layer that prevented the TLS from reconstructing the top of the upper canopy, and that the incomplete tree structure resulted in an underestimation of tree height. Fig. 4a and c demonstrate that a large portion of the TLS was absent in the upper canopy point cloud of the registration result. Dai et al. (2019) and Huo et al. (2022) delineated low vegetation by combining data from TLS and DLS, which suggests ideas for further research to improve the accuracy of tree height estimation.

4.4. Flexible application

In future research, rejecting noise points will be a crucial prerequisite for improving trunk detection accuracy. The results of this study show that both low understory vegetation and high branches can be incorrectly identified as tree trunks, thus overestimating tree detection. Section heights of 3–4 and 4–5 m were determined to be appropriate only for planted cedar forests. For other forest scenarios (such as forest structures with intense understory vegetation or low branch heights), a reasonable threshold could be obtained to minimize overestimation by appropriately increasing or decreasing the slicing height to suit the situation. Another method would be to simultaneously analyze sets of trunk sections and select the section height with the smallest overestimation, for example, the smallest value as a reasonable parameter for detecting tree trunks. Using section thresholds that change with understory vegetation height can also remove understory vegetation points and improve detection accuracy. Further work is required to establish the viability of using different section parameters, such as a decrease in section intervals or the simultaneous processing of multiple layers and layer stacking (Ayrey et al., 2017). Further reduction of the section thickness and consideration of the weight of varying height sections, while ensuring sufficient trunk point cloud density, will refine the processing results and improve the accuracy.

Compared with the traditional CHM-based tree detection method the trunk-based detection method operates directly on the trunk point cloud; while this avoids the effect of surface resolution and smoothing the amount of canopy raster data (McGaughey, 2020), the opportunity to obtain canopy surface raster grid information, which is crucial for single tree segmentation and generation of canopy polygons, is lost.

5. Conclusions

In this study, both DLS and TLS methods were used to reconstruct the point cloud data of a dense plantation forest, including the under-canopy structure. Two LiDAR datasets accurately described the under-canopy structure of 0.3 ha of plantation forest and contained sufficient density trunk point clouds to satisfy the trunk detection requirements. Six datasets of sections at different heights were produced, and the discrete point clouds were transformed into continuous independent projection regions by GIS processing. Finally, the tree locations were expressed by the center coordinates. The results for TLS and manual correction showed that suitable trunk detection parameters were determined for planted cedar forests, which significantly improved tree detection accuracy. This is crucial for assessing forest conditions in dense plantation forests. Estimating forest conditions allow for implementing forest management activities and correctly designating forest policies. Finally, the next research step should focus on setting multiple and variable parameters, as well as exploring the generation of remote-sensing-based models with matching parameters in large-scale mixed forest conditions.

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Authors’ contributions

The authors confirm contribution to the paper as follows: Yupan Zhang and Yuichi Onda, study conception and design; Yupan Zhang and Yiliu Tan, analysis and interpretation of results; Asahi Hashimoto, Chenwei Chiu and Shodai Inokoshi, field investigation and data collection; Yupan Zhang and Yiliu Tan, draft manuscript preparation. All authors reviewed the results and approved the final version of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.fecs.2023.100088.

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